



W H I T E P A P E R

Uncovering User Concerns in a Super-App Ecosystem:

A Topic Modeling Analysis of Gojek App Reviews Using Latent Dirichlet Allocation

Ilham Khadafi

ilhamkhadafi.dkh@gmail.com

github.com/hamkhadafi/gojek-reviews-topic-modeling

ABSTRACT

User-generated reviews on app stores constitute a rich but underutilized source of unstructured feedback. This paper presents an end-to-end topic modeling pipeline applied to 225,000 Indonesian-language reviews of the Gojek super-app (Google Play Store, v4.0.0–v4.9.3). After a multi-stage preprocessing pipeline combining Sastrawi stemming with a domain-specific stopwords lexicon (888 terms), 90,365 reviews (40.2%) were retained for analysis; the 59.8% filtered out consisted predominantly of noise-only content (single-word exclamations, emoji sequences, and sub-three-token fragments) that carry no discriminative topical signal. A 30,000-review sample—whose rating distribution mirrors the retained corpus ($r^2 = 0.99$)—was represented as a Document-Term Matrix ($30,000 \times 3,000$) and modeled using Latent Dirichlet Allocation (LDA). The optimal number of topics was determined through a dual-metric evaluation combining coherence score (C_v) and perplexity used as a convergence diagnostic, with additional qualitative validation via K=6 vs. K=7 topic inspection. The final model (K=7, $C_v = 0.462$) identifies seven semantically distinct topics. Notably, Driver Availability & Order Issues (18.3%) dominates negative reviews, while General Service Satisfaction (17.7%) accounts for 36.6% of five-star reviews. These findings demonstrate the viability of LDA for actionable insight extraction from short-text, informal, multilingual corpora at scale.

Keywords: *topic modeling, LDA, natural language processing, app reviews, Bahasa Indonesia, super-app*

1. Introduction

The proliferation of mobile applications has generated an unprecedented volume of user-generated text. App store reviews represent an organic, real-time feedback channel that contains rich information about user preferences, pain points, and service quality. However, the unstructured nature of this data and its sheer volume render manual analysis intractable at scale.

Gojek, founded in Indonesia in 2010, has grown into one of Southeast Asia's largest super-apps, integrating ride-hailing, food delivery (GoFood), digital payments (GoPay, GoPay Later), and dozens of ancillary services within a single platform. The complexity of this ecosystem means that user reviews span a wide range of service domains, creating a natural use case for automated topic discovery.

Among unsupervised NLP approaches, topic modeling—particularly Latent Dirichlet Allocation (LDA)—has established itself as a standard tool for discovering latent thematic structure in document collections [1]. Its probabilistic generative framework is well-suited to app review data, where a single review may simultaneously reference multiple service aspects. The J-shaped rating distribution common to app store corpora [2]—wherein users review disproportionately at satisfaction extremes—further motivates unsupervised discovery over supervised classification, as ground-truth topic labels are unavailable.

This paper addresses three research objectives:

- Identify the primary topics discussed in Gojek user reviews across a multi-version dataset spanning v4.0.0 to v4.9.3.
- Quantify the distribution of these topics and their relationship to user-assigned star ratings.

- Evaluate model robustness through a dual-metric topic selection procedure and inter-model qualitative comparison.

The results are intended to provide actionable signals for product and operations teams, while demonstrating a reproducible NLP pipeline for short-text, informal Indonesian-language corpora.

2. Dataset

The dataset consists of 225,000 Gojek app reviews sourced from the Kaggle repository “Gojek App Reviews Bahasa Indonesia” [3], covering application versions v4.0.0 through v4.9.3. Five attributes are retained per record.

Table 1. Dataset Attributes

Attribute	Type	Description
userName	String	Display name of the reviewer
content	String	Review text — primary analysis variable
score	Integer (1–5)	Star rating assigned by the reviewer
at	Datetime	Timestamp of the review submission
appVersion	String	Application version at time of review

The rating distribution is heavily bimodal: 5-star reviews constitute 65.4% of the corpus (147,054), while 1-star reviews account for 20.1% (45,229). Reviews in the 2–4 star range collectively represent only 14.5% of the corpus. This J-shaped pattern is consistent with prior characterizations of app store review distributions [2]. The mean review length is 51.4 characters (8.5 words), reflecting the predominantly short-form nature of mobile app reviews.

3. Methodology

The analysis follows a four-stage pipeline: (1) text preprocessing, (2) document-term matrix construction, (3) optimal topic number selection, and (4) LDA model training and interpretation.

3.1 Text Preprocessing

Indonesian app reviews present several preprocessing challenges: informal orthography, extensive abbreviations and slang, code-switching with English, and high lexical variation for the same concept (e.g., “tidak,” “ga,” “gak,” “nggak” all denoting negation). The preprocessing pipeline applies six sequential transformations:

- Lowercasing and removal of URLs, mentions, and hashtags.
- Removal of digits and non-alphabetic characters via regex substitution.
- Removal of tokens shorter than three characters.

- Stopword removal using the union of the Sastrawi stopwords lexicon and a domain-specific custom lexicon (888 terms total). The custom lexicon targets filler words, informal particles, first-person pronouns, and app-brand tokens, while deliberately preserving semantically relevant descriptors such as “cepat” (fast), “mahal” (expensive), and “blokir” (blocked). Custom entries were added iteratively based on inspection of high-frequency tokens that carried no discriminative topical signal.
- Stemming using the Sastrawi morphological analyzer [4], which reduces words to their base form via a rule-based prefix/suffix stripping algorithm optimized for Bahasa Indonesia.
- Post-stem stopwords removal to catch any stopwords produced by the stemmer.

Of the 225,000 original reviews, 90,365 (40.2%) passed the minimum threshold of three meaningful tokens post-preprocessing. The 59.8% filtered out consisted predominantly of noise-only content: single-word exclamations (e.g., “Mantap,” “Oke”), emoji-only entries, and fragments that—after stopwords removal and stemming—yielded fewer than three content-bearing tokens. Such entries carry no discriminative topical signal and their exclusion is unlikely to introduce systematic bias, as they are distributed across all rating tiers.

3.2 Document-Term Matrix

Preprocessed text was converted to a count-based Document-Term Matrix (DTM) using scikit-learn’s CountVectorizer [5]. Parameter choices reflect standard heuristics for informal short-text corpora:

- `max_df = 0.90`: removes tokens appearing in more than 90% of documents, as near-universal tokens carry no discriminative value across topics.
- `min_df = 10`: removes tokens appearing in fewer than 10 documents, filtering likely noise and misspellings. This threshold was selected to balance vocabulary coverage against noise, consistent with practice in comparable short-text studies.
- `max_features = 3,000`: caps the vocabulary to the 3,000 most frequent tokens post-filter, following the recommendation in Blei et al. [1] to constrain vocabulary for efficiency without significant information loss at this corpus scale.
- `ngram_range = (1, 2)`: includes unigrams and bigrams to capture compound expressions (e.g., “gopay later,” “terima kasih”).

A random sample of 30,000 reviews was drawn from the 90,365 retained reviews for model training. The sample’s rating distribution (1★: 40.9%, 2★: 8.3%, 3★: 7.1%, 4★: 6.4%, 5★: 37.3%) closely mirrors that of the full retained corpus ($r^2 = 0.99$), confirming representativeness. Full-corpus training at this vocabulary size was computationally prohibitive given the quadratic scaling of LDA’s memory footprint with vocabulary size and document count; 30,000 documents provides sufficient statistical coverage for stable topic estimation at $K \leq 7$ [1]. The resulting DTM has shape $30,000 \times 3,000$ with a sparsity of 0.9975.

3.3 Optimal Topic Number Selection

Selection of the number of topics K is a critical step in LDA. This analysis employs a dual-metric strategy:

Coherence Score (C_v) measures the average semantic similarity between the top words of each topic, using a sliding window over the corpus and normalized pointwise mutual information (NPMI). C_v has been shown to correlate well with human topic interpretability judgments [6] and serves as

the primary selection criterion here. For reference, C_v values above 0.40 are generally considered indicative of interpretable topics in short-text corpora [6]; the selected model achieves $C_v = 0.462$, placing it in the acceptable-to-good range for this domain.

Perplexity is retained as a convergence diagnostic—confirming that models trained at each K have reached a stable solution—rather than as a K selection criterion. Perplexity in scikit-learn’s online variational Bayes implementation increases monotonically with K due to approximations inherent to the algorithm [5], making cross- K comparison uninformative for model selection purposes.

K was evaluated over the set $\{3, 4, 5, 6, 7, 8, 10\}$, with all models trained using identical hyperparameters to ensure comparability.

Table 2. K Selection Results

K	Perplexity (diagnostic)	Coherence C_v (primary)	Notes
3	913.65	0.420	Too few topics for a super-app
4	990.40	0.321	Coherence drop; topics merge
5	1011.20	0.365	Baseline (prior work)
6	993.28	0.411	Candidate; compared against $K=7$
7 ★	983.60	0.462 ★	Selected — highest C_v ; qualitatively validated
8	1016.56	0.460	Near-identical to $K=7$; marginal gain
10	1058.23	0.403	Coherence declines; over-segmentation

Coherence score peaked at $K=7$ ($C_v=0.462$), with $K=8$ scoring near-identically (0.460). To adjudicate between $K=6$ and $K=7$, a qualitative topic inspection was conducted. $K=6$ conflates satisfaction signals with account management keywords in a single topic, while $K=7$ produces a clean separation between GoPay wallet issues (T4) and GoPay Later payment complaints (T6)—a distinction with direct operational implications. Additionally, $K=7$ separately identifies driver availability friction (T0) and in-trip delivery experience (T5), two operationally distinct failure modes. The mean review length of 8.5 words was considered as a potential constraint on topic separability; topic inspection confirmed that $K=7$ remains interpretable at this text length. $K=7$ was therefore selected as the final model.

3.4 LDA Model

LDA models each document d as a mixture of K topics, where each topic k is a distribution over vocabulary V . The generative process is governed by [1]:

$$p(\beta, \theta, z, w) = \prod_k p(\beta_k | \eta) \cdot \prod_d p(\theta_d | \alpha) \cdot \prod_n p(z_{dn} | \theta_d) \cdot p(w_{dn} | \beta_{1:k}, z_{dn})$$

where α is the Dirichlet prior on per-document topic distributions and η is the prior on per-topic word distributions. Inference was performed using online variational Bayes (`learning_method='online'`), which processes documents in mini-batches and scales efficiently to large corpora.

Hyperparameters were set as follows: `max_iter=30` (sufficient for convergence as confirmed by perplexity stabilization across runs); `learning_decay=0.7` (the default recommended value in Hoffman et al. [7] for online VB, controlling the rate at which earlier gradient estimates are down-weighted); `max_iter` was held constant across all K evaluations to ensure apple-to-apple comparability. `random_state=42` was fixed throughout for full reproducibility.

4. Results

4.1 Identified Topics

The K=7 LDA model identifies seven semantically distinct topics. Table 3 presents each topic's label, top keywords, document share, and median topic assignment probability.

Table 3. Identified Topics (K=7 LDA Model, n=30,000)

ID	Label	Top Keywords	Share	Median Prob.
T0	Driver Availability & Order Issues	driver, order, susah, voucher, buka, gofood, dapet, nunggu, cari, jam	18.3%	0.512
T1	GoFood Pricing & Promotions	makan, mahal, biaya, promo, harga, gofood, diskon, ongkir, bikin, beli	14.2%	0.518
T2	General Service Satisfaction	bantu, kasih, mudah, bagus, bintang, terima, layan, cepat, moga, parah	17.7%	0.536
T3	GoRide Pricing & Account Management	ganti, gimana, pakai, bantu, murah, goride, nomor, suka, tarif, muncul	9.0%	0.440
T4	GoPay Wallet & Transactions	gopay, saldo, akun, transaksi, top, kirim, lapor, upgrade, daftar, uang	13.9%	0.516
T5	Driver & Delivery Experience	driver, drivernya, pesan, jalan, layan, jemput, waktu, sesuai, pilih, lokasi	13.6%	0.478
T6	Payment & GoPay Later	bayar, gopay, pake, pakai, ramah, blokir, akun, gopaylater, sistem, paylater	13.5%	0.524

Topic distribution is relatively balanced, ranging from 9.0% (T3) to 18.3% (T0). Topic T3 (GoRide Pricing & Account Management) warrants particular attention: it is the smallest topic (9.0%) and exhibits the lowest median assignment probability (0.440) with the widest interquartile range. This reflects genuine thematic breadth rather than a modeling artifact—GoRide users frequently co-reference pricing concerns alongside account access issues (e.g., verification failures, number changes) within the same review, creating a naturally heterogeneous topic. Representative high-

probability reviews confirm this co-occurrence pattern. Nonetheless, the lower assignment confidence signals that T3 should be interpreted with more caution than other topics.

4.2 Topic-Rating Relationship

Cross-tabulating dominant topic assignments against star ratings reveals strong, interpretable associations:

- T2 (General Service Satisfaction) accounts for 36.6% of all 5-star reviews—more than double its overall share (17.7%)—confirming that this topic captures genuine positive sentiment rather than generic language.
- T0 (Driver Availability & Order Issues) is the leading topic in both 1-star (23.4%) and 2-star (25.4%) reviews, identifying driver-related operational failures as the primary driver of severe dissatisfaction.
- T4 (GoPay Wallet & Transactions) appears in 20.2% of 1-star reviews, disproportionate to its 13.9% overall share, indicating that financial transaction failures generate outsized negative reactions.
- T1 (GoFood Pricing & Promotions) distributes relatively evenly across all ratings (11.0%–17.1%), suggesting pricing discussions occur across the satisfaction spectrum—both as complaints and as acknowledgment of promotions.
- T3 (GoRide) shows the most uniform distribution across ratings, consistent with its mixed thematic content.

4.3 Model Confidence

Topic assignment probability—the maximum posterior probability assigned to a document's dominant topic—serves as a model confidence measure. Median probabilities range from 0.440 (T3) to 0.536 (T2). Topics T3 and T5 exhibit the lowest medians and widest interquartile ranges, consistent with their semantic proximity: both reference driver services, and documents in either topic frequently contain tokens from the other. This cross-topic overlap is an expected property of LDA applied to related service domains. The overall moderate confidence level (all medians between 0.44–0.54) is consistent with short-text LDA behavior, where limited token count per document constrains posterior concentration.

5. Discussion

5.1 Business Implications

The topic structure maps directly onto Gojek's major service verticals, enabling targeted intervention:

- Driver operations (T0 + T5, combined 31.9%) constitute the largest aggregate concern area. The separation between availability issues (T0: difficulty finding drivers, order cancellations, voucher conflicts) and delivery experience (T5: map accuracy, pickup coordination, route

adherence) is operationally meaningful—these require different remediation strategies: supply-side driver allocation versus navigation UI improvements.

- Financial services (T4 + T6, combined 27.4%) represent the second-largest concern area. GoPay wallet failures (T4: failed transfers, saldo discrepancies) and GoPay Later billing issues (T6: unexpected charges, payment due date disputes) are distinct user problems that share the attribute of high emotional salience—users react more severely to perceived financial failures than to service inconveniences.
- Pricing sensitivity (T1, 14.2%) reflects user awareness of delivery fee structures. The even distribution across rating tiers suggests that promotional pricing exceeding expectations generates positive reviews, while fee increases generate negative ones—making this topic a useful signal for pricing strategy monitoring.
- The satisfaction topic (T2, 17.7%) functions as a baseline for measuring overall service perception and can serve as a version-level tracking metric.

5.2 Limitations

Several methodological limitations should be noted:

- LDA assumes a bag-of-words representation, discarding word order and syntactic context. Negation handling is a known weakness: “tidak bagus” (not good) may be parsed as “bagus” after stopword removal, inverting semantic polarity.
- The short average review length (8.5 words) limits topic separability and contributes to the moderate median assignment probabilities observed across all topics.
- Minor vocabulary overlap between T4 and T6 (both reference ‘gopay’ and ‘akun’) is an inherent consequence of shared service infrastructure. The distinction is preserved at the level of co-occurring terms but may not generalize to all edge cases.
- Sastrawi’s rule-based stemmer does not handle all informal Indonesian variants (e.g., ‘ongkirnya’ is not reduced to ‘ongkir’), creating vocabulary fragmentation for colloquial terms.
- The analysis captures a static snapshot of user sentiment. Temporal dynamics across app versions are not modeled.

6. Conclusion and Future Work

This paper demonstrates an end-to-end LDA pipeline for extracting actionable topic structure from 225,000 informal Indonesian-language app reviews. The dual-metric K selection procedure—combining coherence score (C_v) as the primary criterion with perplexity as a convergence diagnostic, supplemented by qualitative inter-model comparison—yields a seven-topic model that is both statistically motivated and domain-interpretable. Key findings include the primacy of driver-related operational issues as the dominant complaint category, the strong association between a general satisfaction topic and five-star reviews, and the disproportionate negative impact of financial transaction failures on user sentiment.

Future work could extend this analysis in three directions: (1) replacing LDA with transformer-based topic models such as BERTopic with IndoBERT embeddings [8] to capture richer semantic context and address LDA’s bag-of-words limitation; (2) conducting longitudinal topic analysis across app versions to track how user concerns evolve with product changes; and (3) integrating topic assignments with aspect-based sentiment analysis that explicitly handles negation, rather than

standard polarity scoring, to enable topic-level sentiment profiling robust to the linguistic patterns characteristic of informal Indonesian reviews.

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